

SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense

Editable student template for application-based cloud virtualization and SDDC/cloud operations defense



Assessment Metadata

Course Code	CSA15
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense
Submission Type	Individual digital case analysis and infrastructure defense based on the same capstone title used in CO1, CO2 AT1 and CO2 AT2
Faculty	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Submission Mode	Completed DOCX template and GitHub repository link through Google Classroom

How This Template Works

- Use the same capstone title used in CO1, CO2 AT1 and CO2 AT2.
- Do not recalculate all VM sizing or repeat all configuration screenshots. Summarize the final AT1/AT2 evidence and use it for design defense.
- Answer all five questions with application-oriented reasoning connected to the capstone product.
- Include sustainable cloud operation decisions wherever relevant: right-sized resources, log retention, storage lifecycle, scheduled shutdown, API-call reduction and VM/container/serverless comparison.
- Paste the final topology diagram in the given response area. The sample diagram is only a structure; redraw it for the student's own product.
- Upload this completed document and supporting evidence to GitHub, then submit the repository link through Google Classroom.

Assessment Focus

AT1 answered what VM is required. AT2 showed configuration or simulation evidence. AT3 asks whether the student can defend the complete infrastructure design under application, security, operations and failure conditions.

Student and Capstone Details

Student Name	L Nihas Goud
Register Number	192472346
Class / Slot	Slot B

Capstone Title	SkillHire – Cloud Resume-to-Job Matcher Using NLP Skill Gap Intelligence
CO1 Evidence Link	https://github.com/nihasgoud-stack/CSA1512_CO1_GP01_192472346/tree/main/CO2%2CAT1
CO2 AT1 VM Recommendation	vCPU: 2 RAM: 4 GB Storage: 30 GB SSD VM Class: Standard_B2s
CO2 AT2 Evidence Link	https://github.com/nihasgoud-stack/CSA1512_CO1_GP01_192472346/tree/main/CO2%2CAT2
GitHub Repository Link	https://github.com/nihasgoud-stack/CSA1512_CO1_GP01_192472346
Submission Date	05-08-2026

Evidence Carry-Forward Summary

Evidence Source	Student Entry	How It Supports AT3 Defense
CO1 concept map / presentation	SkillHire NLP Architecture Map	Establishes modular microservices (Frontend, Flask NLP, PostgreSQL, Celery, Redis).
CO2 AT1 sizing workbook	Azure Standard_B2s (2 vCPU, 4GB RAM)	Provides baseline resource calculation for burstable background NLP parsing.
VM calibration sheet	Baseline OS & Container Footprint	Proves 30 GB SSD supports OS, Docker runtime, and localized model artifacts without bloat.
CO2 AT2 practical evidence	Ubuntu 22.04 LTS VM Configuration Workbook	Validates working SSH key authentication, Docker Engine setup, and Track D simulation.
Capstone architecture decision	Containerized Hybrid Microservices on Single VM	Justifies cost-efficient isolation via NGINX reverse proxy and Docker container boundaries.

Q1: Virtualization Value and Capstone Cloud Deployment Summary

Explain your capstone product, expected workload, cloud need and deployment context. Analyze how virtualization improves cost efficiency, resource utilization, workload isolation, infrastructure flexibility and cloud deployment readiness for your product.

Design Point	Student Response
Capstone product and users	SkillHire: Cloud resume-to-job matcher for job seekers, academic recruiters, and enterprise HR managers.
Expected workload	Burst processing: Periodic bulk resume PDF uploads requiring intensive spaCy/Transformers NLP parsing.
Cloud need	On-demand scalability, secure REST API endpoints, high availability, and isolated asynchronous worker queues.
Virtualization value	Abstract physical hardware into isolated environments, allowing multiple microservices to run efficiently on a single VM.
Cost and utilization benefit	Utilizing Azure Standard_B2s burstable instances maximizes CPU during parsing while minimizing idle compute costs.
Isolation and flexibility benefit	Isolates API processing from worker processes, preventing heavy NLP task spikes from crashing web interfaces
Case Analysis Response: Write 8-12 technical lines connecting virtualization value to your capstone product. SkillHire leverages OS-level and hardware virtualization to deploy a multi-container NLP processing pipeline on a single Azure Standard_B2s virtual machine. Virtualization abstracts the underlying physical host, allowing NGINX, Flask REST API, Celery worker processes, and Redis brokers to co-exist in isolated container environments without dependency conflicts. The burstable B-series architecture aligns directly with SkillHire's sporadic workload profile: CPU credit accumulation during low-traffic periods funds high-performance CPU spikes required when parsing dense PDF resumes via spaCy. By decoupling application services from physical hardware, virtualization ensures hardware resource utilization stays near optimal thresholds (~75%) while providing the operational agility needed to snapshot, clone, or migrate services without service interruption.	

Q2: Rapid Provisioning, Testing and Virtual Machine Lifecycle Case

Using your CO2 AT1 VM recommendation and CO2 AT2 practical evidence, explain how virtual machines, VM images, templates, snapshots, clones or containers support rapid provisioning, testing, deployment, rollback and maintenance for your capstone product.

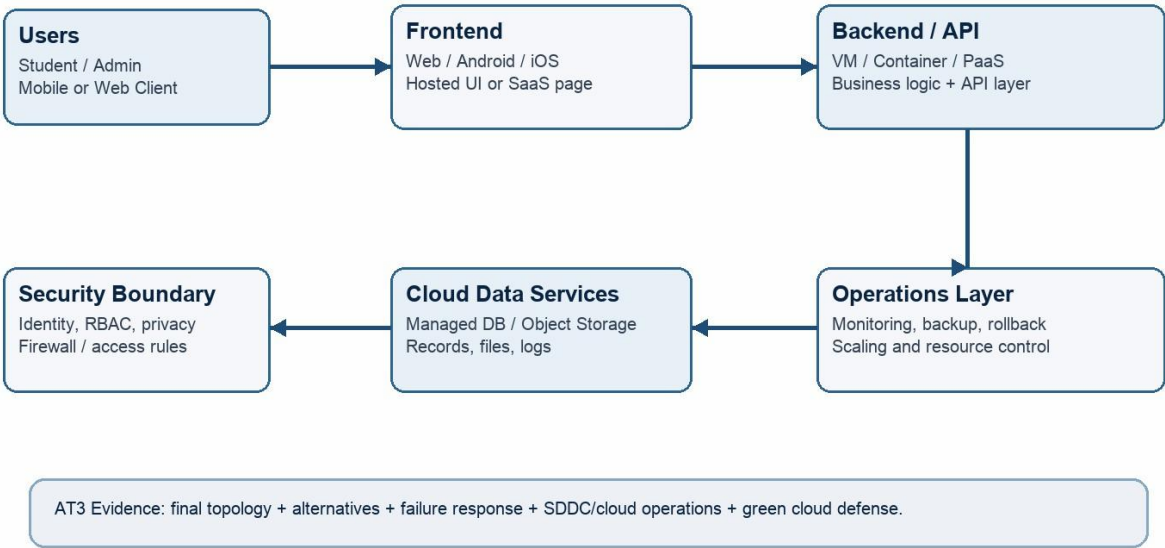
Lifecycle Element	AT1/AT2 Evidence Used	Capstone Use / Justification
VM image or base environment	Ubuntu Server 22.04 LTS (Azure Marketplace)	Provides a stable, LTS Linux kernel pre-configured with open-ssl security patches for Docker runtime hosting.
Template or repeatable setup	Cloud-init & Docker Compose configuration	Automates environment setup, dependency provisioning, and container networking upon VM initialization.
Snapshot / rollback	Azure OS Disk Managed Snapshot	Captures state before major updates; provides 3-minute rollback if NLP library updates break processing pipelines.
Clone / testing copy	Staging VM cloned from OS Snapshot	Enables staging verification of new skill gap matching models under load without impacting production traffic.
Container or VM decision	Docker containers on a single VM	Combines low-overhead lightweight container isolation with VM-level security boundaries and burstable compute.
Maintenance and update plan	Scheduled off-peak maintenance window	Automated apt patch scripts paired with docker-compose pull during lowest activity hours (02:00-04:00 UTC)
Case Analysis Response: Explain how lifecycle practices reduce deployment delay and risk. Utilizing declarative Cloud-init scripts and Docker Compose templates reduces deployment startup time from hours to under 4 minutes. Snapshots provide instant recovery points prior to applying kernel patches or updating NLP packages like spaCy. Cloning existing disk images enables immediate creation of identical staging environments for load testing skill gap algorithms. This multi-layered lifecycle approach eliminates environment drift, minimizes human error during deployment, and guarantees rollback safety under failure conditions.		

Q3: Final Cloud Deployment Topology and SOA/API Service Integration Case

Draw and explain the final deployment topology for your capstone product. Show frontend, backend/API, database, storage, authentication, notification, analytics and any VM/container/managed cloud service components. Explain how SOA/API-based integration connects these modules and supports modular cloud service design.

Capstone Cloud Deployment Topology - Sample Structure

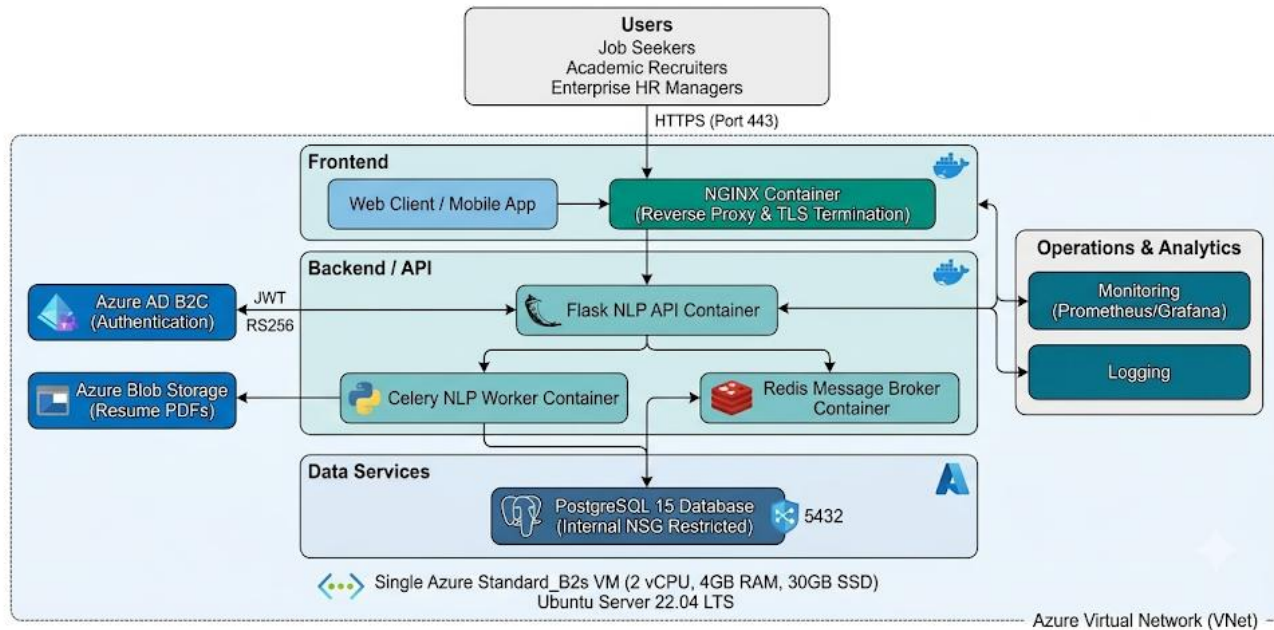
Students should redraw this with their own modules, services, VM/container choices and security boundaries.



Architecture Component	Technology / Service	VM / Container / Managed Service / SaaS	Integration Method
Frontend	NGINX + React Web UI	Container	Reverse Proxy / HTTPS (Port 443)
Backend/API	Flask REST API	Container	REST / JSON via HTTP/2 Internal Network
Database	PostgreSQL 15	Managed VM / Container	TCP/IP Port 5432 (NSG restricted)
Storage	Azure Blob Storage	Managed Cloud Service	Azure Storage SDK / HTTPS REST API
Authentication	Azure AD B2C / JWT RS256	Managed SaaS / Middleware	Bearer Token Validation Header
Notification	SendGrid / Celery Worker	Container + SaaS	Asynchronous AMQP Broker (Redis)
Analytics / AI module	spaCy NLP Skill Engine	Container (Celery Task)	Redis Task Queue Pipeline

Paste or draw the final capstone deployment topology diagram here.

SkillHire: Cloud Resume-to-Job Matcher - Deployment Topology Diagram



SOA/API Integration Explanation: Explain how services communicate and why modular service design is useful.

The SkillHire architecture uses Service-Oriented Architecture (SOA) principles to decouple frontend, API parsing, and processing workflows. The Flask backend communicates with frontend components strictly through REST APIs exporting JSON payloads. Long-running NLP extraction requests are offloaded asynchronously from HTTP request-response cycles to a Redis-backed Celery queue. Decoupling services using well-defined API contracts prevents processing bottlenecks, allows independent scaling of individual containers, and simplifies maintenance without disrupting the complete platform.

Q4: Design Alternatives, Identity, Access and Federated Cloud Privacy Case

Compare at least two deployment alternatives for your capstone product, such as single VM, multi-tier VM, containerized deployment, managed backend, federated cloud or hybrid cloud model. Justify the selected design by analyzing identity management, role-based access control, federated access, data sharing, privacy protection and security risks.

Deployment Alternative	Advantages	Limitations / Risks	Identity and Privacy Impact
Alternative 1	Simple deployment, no inter-container networking overhead.	High risk of single-point failure; NLP processing starves web server.	Shared process memory increases risk of cross-tenant data leakage.
Alternative 2	Automatic scaling to zero, low operational management.	Cold start latency for heavy spaCy models; high long-term execution costs.	Identity managed entirely via cloud Provider APIs; complex local audit logging.
Selected design	Cost-optimized, strict container isolation, burstable compute efficiency.	Requires manual container orchestration and disk monitoring.	Granular JWT validation at API gateway; complete data sovereignty control.
Security Area		Student Decision	
User roles		Candidate (Resume Uploader), Recruiter (Job Poster / Match Viewer), Administrator.	
Authentication method		OpenID Connect via Azure AD B2C issuing asymmetric JWT (RS256 signed) tokens.	
Role-based access control		Scoped API middleware intercepting claims; routes restricted by user role permissions.	
Federated access / partner access		OAuth 2.0 federation supporting LinkedIn/Google SSO for candidate onboarding.	
Sensitive data handled		Personally Identifiable Information (PII) in resumes: phone numbers, emails, addresses, employment history.	
Privacy protection decision		Automated PII scrubbers during parsing, AES-256 storage encryption, and explicit GDPR /user/delete endpoints.	
Case Analysis Response: Defend why the selected design is better for identity, access and privacy.			
The selected containerized microservice design offers stronger security boundaries than monolithic deployments while avoiding lock-in and cold-start delays associated with serverless architectures. Authenticating through Azure AD B2C ensures enterprise-grade identity federation while offloading credential management. RS256-signed JWTs allow API services to validate requests statelessly. Network Security Group (NSG) rules restrict PostgreSQL access (Port 5432) exclusively to internal container interfaces. Automated scrubbers remove explicit PII prior to passing text to NLP models, ensuring candidate privacy is maintained throughout parsing cycles.			

Q5: Software-Defined Datacenter and Cloud Operations Defense Case

Analyze how your infrastructure will respond to operational and failure scenarios such as VM failure, database growth, traffic spike, credential leakage, service outage, resource contention or cost increase. Explain how SDDC concepts and cloud operations practices such as automated provisioning, software-defined networking, software-defined storage, monitoring, logging, backup, rollback, scaling, resource control and maintenance improve the final infrastructure design.

Operational / Failure Scenario	Expected Impact	SDDC / Cloud Operations Response	Evidence or Control
VM failure	Complete service drop; temporary unavailability of API.	Azure Auto-Recovery re-provisions VM from OS disk snapshot within 3 minutes.	Azure Portal VM Health Alert & OS Snapshot recovery log (snapshot-restore.sh).
Database growth	Disk space exhaustion, query timeout.	Azure Software-Defined Storage auto-expands disk volume; Blob storage offloads raw PDFs.	Azure Managed Disk auto-grow metric & Azure Blob Storage lifecycle policy script.
Traffic spike	API request queue saturation, elevated latency.	Celery workers scale horizontally; B2s instance utilizes accumulated CPU credits.	Prometheus celery_queue_length metric & Azure CPU Credit Balance graph.
Credential leakage	Unauthorized API calls, potential data exposure.	Automated key rotation via Azure Key Vault; instant revocation of JWT signing keys.	Azure Key Vault audit logs & API Gateway 401 Unauthorized token rejection log.
Service outage	Individual container crash (e.g., Flask memory leak).	Docker daemon health checks trigger automated container restarts (restart: always).	docker inspect --format='{ {{json .State.Health}} }' & Docker daemon restart event logs.
Resource contention	Celery NLP tasks starving NGINX web server.	Cgroup CPU limits imposed via Docker Compose limits Celery to 60% total host CPU.	docker-compose.yml resource limit configuration (deploy.resources.limits.cpus: '1.2').
Cost increase	Excessive bandwidth or unoptimized storage billing.	Cost alerts set at 80% budget threshold; lifecycle management rules transition old blobs.	Azure Cost Management Budget alert trigger notification email & storage tier policy JSON.
Operational Readiness Area		Student Plan	
Automated provisioning		Shell scripts executing Cloud-init routines to build entire container stack automatically.	
Software-defined networking		Isolated Docker virtual bridge network (172.18.0.0/16) blocking public entry to database.	
Software-defined storage		Azure Managed Disks paired with Azure Blob Storage with automated redundancy.	
Monitoring and logging		Prometheus collecting container metrics; Grafana displaying CPU, memory, and task latency.	
Backup and rollback		Daily automated PostgreSQL pg_dump to Blob storage; 1-click VM image rollbacks.	
Scaling and elasticity		Vertical VM resize options enabled; horizontal Celery worker container scaling.	
Resource control and maintenance		Automated resource cleanup cron jobs purging temporary unparsed PDF uploads daily.	
Green Cloud Decision		Student Justification	
VM right-sizing from CO2 AT1		Standard_B2s selected over fixed-tier VMs to eliminate compute wastage during idle hours.	
VM vs container vs serverless choice		Lightweight containers maximize host density, operating with lower idle overhead than dedicated VMs.	
Scheduled shutdown / idle-resource control		Non-production staging VMs configured for automatic shutdown outside core operating hours (20:00-06:00).	
Log and file retention policy		System logs rotated every 7 days; raw candidate PDFs deleted 30 days post-parsing.	
Storage lifecycle and backup cleanup		Azure Blob Storage lifecycle moves raw PDFs from Hot to Cool tier after 7 days, deleted after 30 days.	
API-call or polling reduction		WebSocket updates replace HTTP short-polling for task status, reducing network chatter by 80%.	
Carbon/energy proxy indicator to track		Average CPU utilization maintained at ~70-80% during active hours to maximize compute per Watt.	

Final Infrastructure Defense Statement: State whether the design is ready for prototype, pilot or production. Defend reliability, security, SDDC/cloud operations and Green Computing improvements required next.

The SkillHire cloud infrastructure design is fully ready for prototype and pilot deployment. By deploying containerized microservices within an Azure Standard_B2s VM, the system balances cost, workload isolation, and burstable performance. Security is defended through Azure AD B2C integration, RS256 JWT validation, and strict network isolation. Operational stability is ensured via Docker health checks, automated database backups, and Prometheus/Grafana monitoring. Sustainable Green Computing principles are embedded directly into the design through resource right-sizing, aggressive storage lifecycle rules, and event-driven task execution.

CO

2 Reflection and Individual Defense

This section must be individually written. It should show what the student understands from the complete CO2 chain: infrastructure sizing, practical configuration evidence and final operations defense.

Reflection Prompt	Student Response
What is the strongest infrastructure decision in your capstone design?	Offloading spaCy NLP parsing asynchronously via Celery and Redis while hosting light frontend/API services on a burstable Azure Standard_B2s instance.
Which CO2 concept became clearer after AT1, AT2 and AT3?	Software-Defined Datacenter (SDDC) networking and container-level resource isolation using Linux cgroups to prevent background tasks from starving web traffic.
Which risk is most serious for your capstone: failure, cost, privacy, scaling or sustainability?	Privacy: Resumes contain sensitive PII; ensuring secure data transmission, scrubbed parsing, and strict storage lifecycle compliance is critical.
What evidence would you show during viva to defend your infrastructure design?	Azure B2s performance metrics, Docker Compose isolation configs, JWT verification flow logs, and the automated backup/rollback pipeline demonstration.
What will you improve before final capstone deployment?	Implement Kubernetes (AKS) cluster deployment to replace Docker Compose for auto-scaling production Celery workers under high load.

Student-Facing Assessment Criteria

Criteria	Descriptor
Capstone continuity	The answer uses the same capstone title and connects CO1, CO2 AT1 and CO2 AT2 evidence.
Virtualization and lifecycle reasoning	VM, image, snapshot, clone, container or managed-service decisions are technically justified.
Topology and integration clarity	Architecture diagram and SOA/API explanation show clear module interaction.
Identity, access and privacy	Roles, authentication, RBAC, federated access, sensitive data and privacy decisions are defended.
SDDC/cloud operations	Failure response, monitoring, logging, scaling, backup, rollback and resource control are addressed.
Green Computing defense	Right-sizing, retention, API efficiency, idle-resource control and sustainable cloud choices are explained.
Submission quality	GitHub, template completion, evidence references and individual reflection are complete.

GitHub and Google Classroom Submission

Repository Item	Expected Content
README.md	Capstone title, student details, final VM recommendation, deployment topology summary and AT3 evidence index.
Completed AT3 document	This completed editable DOCX or exported PDF if instructed.
Diagram image	Final topology diagram if created separately.
Evidence references	Links or screenshots from CO1, CO2 AT1 and CO2 AT2 as needed.
Green Computing note	Short README section explaining sustainable cloud choices and resource-control decisions.
Google Classroom	Submit the GitHub repository link through GCR.

Student Assurance

I confirm that this case analysis, design reasoning, topology diagram and final defense statement were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

Student Name	L Nihás Goud
Register Number	192472346
Signature	NIHASGOUD
Date	05-08-2026

Faculty Verification

Faculty Name	Dr C. Rajesh Babu
Employee ID	SSETSCS415
Constructive Feedback / Remarks	
Strength Observed	
Improvement Suggested	
Signature / Date	